

CSE360
Final Exam

1.(same type maybe)

Question 12.4.

A microprocessor provides an instruction capable of moving a string of bytes from one area of memory to another. The fetching and initial decoding of the instruction takes 10 clock cycles. Thereafter, it takes 15 clock cycles to transfer each byte. The microprocessor is clocked at a rate of 10 GHz.

- a) Determine the length of the instruction cycle for the case of a string of 64 bytes.
- b) What is the worst-case delay for acknowledging an interrupt if the instruction is non interruptible?
- c) Repeat part (b) assuming the instruction can be interrupted at the beginning of each byte transfer.

2.

Problem-04:

A disk pack has 19 surfaces and storage area on each surface has an outer diameter of 33 cm and inner diameter of 22 cm. The maximum recording storage density on any track is 200 bits/cm and minimum spacing between tracks is 0.25 mm. Calculate the capacity of disk pack.

3.

Problem: 7.17

A 32-bit computer has two selector channels and one multiplexor channel. Each selector channel supports two magnetic disks and two magnetic tape units. The multiplexor channel has two-line printers, two card readers, and 10 VDT terminals connected to it. Assume the following transfer rate:

Disk drive: 800 Kbytes/sec

Magnetic tape drive 200 Kbytes/sec

Line printer 6.6 Kbytes /sec

Card reader: 1.2 Kbytes/sec

VDT: 1 Kbytes/sec

Estimate the maximum aggregate I/O transfer rate in this system.

Selector selects one device at a time, Multiplexer for getting output from one or more than one device.

For Selector I/O transfer rate = maximum of (two magnetic disk and two magnetic tape units) = $\max(800, 200) = 800 \text{ KBps}$

For two selectors = $800 * 2 = 1600 \text{ KBps}$

For Multiplexer I/O transfer rate = $2 * 6.6 + 2 * 1.2 + 10 * 1 = 25.6 \text{ Kbyte/sec}$

So, total maximum aggregate I/O transfer rate = $1600 + 25.6 = 1625.6 \text{ Kbyte/sec}$

4.

It takes a machi to fetch instruction 7 clock cycle. 1 byte transfer takes 13 cycle. Clock rate 32ghz. Whats the time take to transfer 512 byte?

5.

Use of Stack for computation

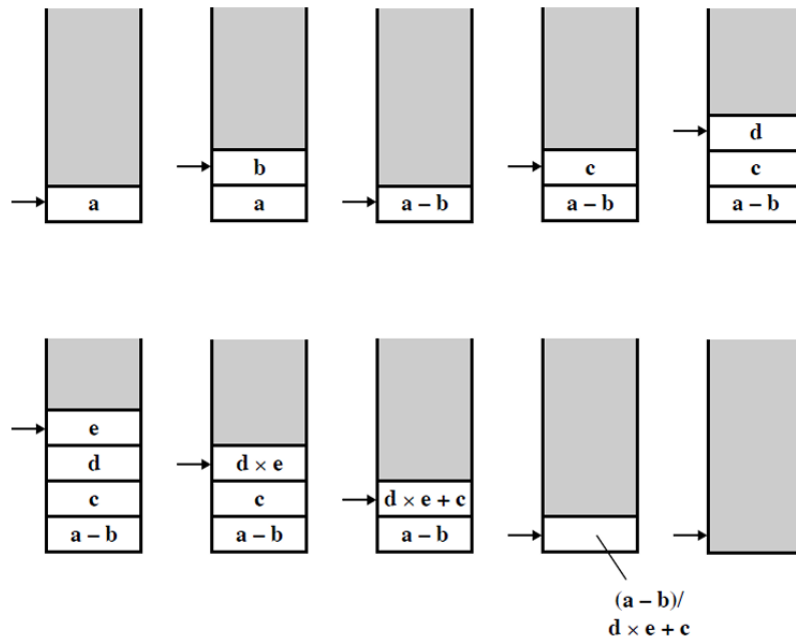


Figure 10.16 Use of Stack to Compute $f = \frac{a - b}{c + (d * e)}$

6.

Question 11.4

Consider a 16-bit processor in which the following appears in main memory, starting at location 200:

200	Load to AC	Mode
201	500	
202	Next instruction	

The first part of the first word indicates that this instruction loads a value into an accumulator. The Mode field specifies an addressing mode and, if appropriate, indicates a source register; assume that when used, the source register is R1, which has a value of 400. There is also a base register that contains the value 100. The value of 500 in location 201 may be part of the address calculation. Assume that location 399 contains the value 999, location 400 contains the value 1000, and so on. Determine the effective address and the operand to be loaded for the following addressing modes.

- Immediate
- Direct
- Indirect
- PC relative
- Displacement
- Register
- Register indirect
- Autoindexing with increment, using R1

No.	EA	Operand
A	201	500
B	500	1100
C	1100	1700
d	702	1302
e	600	1200
f	R1	400
g	400	1000
h	400	1000

4. Analyze and compare zero-, one-, two-, and three- address machines by writing programs to compute the following expression for each of the four machines.
 $(A+B/C) - (D + E) + F \cdot G$

The instructions available for use are as follows:

0 Address	1 Address	2 Address	3 Address
PUSH M	LOAD M	MOV ($X \leftarrow Y$)	MOVE ($X \leftarrow Y$)
POP M	STORE M	ADD($X \leftarrow X+Y$)	ADD($X \leftarrow Y+Z$)
ADD	ADD M	SUB ($X \leftarrow X-Y$)	SUB($X \leftarrow Y-Z$)
SUB	SUB M	MUL ($X \leftarrow X * Y$)	MUL($X \leftarrow Y * Z$)
MUL	MUL M	DIV($X \leftarrow X/Y$)	DIV($X \leftarrow Y/Z$)
DIV	DIV M		

8.

8. Consider a four-stage pipeline processor. The number of cycles needed by the four instructions namely I1, I2, I3, and I4 for the stages S1, S2, S3, and S4 are respectively given in the table below. Show the diagram of instructions executions and calculate the number of cycles needed to execute the instructions.

	S1	S2	S3	S4
I1	2	1	1	2
I2	1	2	2	2
I3	2	2	1	3
I4	1	3	2	2